
HARNESSING THE POWER OF CNNs FOR AUTOMATED SKIN CANCER CLASSIFICATION

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Brief Project Description

Skin cancer is among the most common malignancies worldwide, with over 80,000 cases reported in Canada each year [1]. Identifying it at an early stage greatly improves patient outcomes, with five-year survival rates reaching up to 99% [2]. However, limited access to dermatologists—particularly in rural areas—creates barriers to timely diagnosis, and visual inspection alone yields only about 60% accuracy [3]. This project aims to develop an automated deep learning system that classifies dermoscopic images into seven lesion categories using the HAM10000 dataset [4]. The system assists dermatologists by providing fast, reliable second opinions that can improve diagnostic accuracy and reduce clinical workload. Deep learning, specifically convolutional neural networks (CNNs), can automatically learn discriminative visual features such as color distribution, texture, and border irregularities essential for skin-lesion assessment [5]. Unlike traditional methods requiring manual feature extraction, CNNs learn hierarchical patterns directly from raw images, making them ideal for modeling the complex variability in dermoscopic data.



Figure 1: Pipeline for skin lesion classification.

As shown in Figure 1, the pipeline takes a dermoscopic image as input, applies preprocessing and data augmentation to enhance robustness, and feeds it into a fine-tuned ResNet-18 model that outputs the predicted lesion category. This automated classification framework demonstrates the potential of deep learning to support early skin-cancer detection, especially in regions with limited access to dermatologists.

Data Processing

The HAM10000 dataset consists of 10,015 dermoscopic images spanning seven skin lesion categories. The class distribution is shown in Figure 2. Data cleaning involved removing corrupted, duplicate, and incomplete samples to ensure high-quality inputs. All images were resized to 224×224 pixels and normalized to ensure consistent input for the model. The held-out 15% test set, containing unseen and unaugmented images, was reserved for unbiased performance evaluation. To further assess model generalization, a small set of external images not included in HAM10000 will be used for final testing. Since the nv class dominated the dataset, leading to class imbalance and reduced accuracy on rarer lesions, minority classes were oversampled using data augmentation, resulting in a balanced distribution as displayed in Figure 3. One key challenge during preprocessing was managing

class imbalance without introducing overfitting, as excessive augmentation risked generating visually redundant samples.

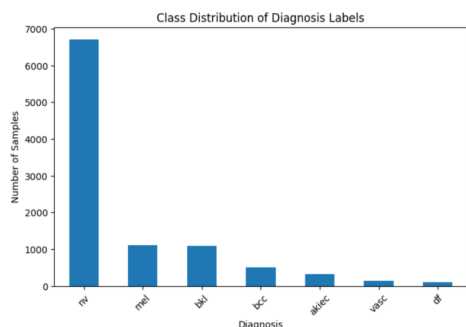


Figure 2: Distribution of skin lesion classes in the HAM10000 dataset.

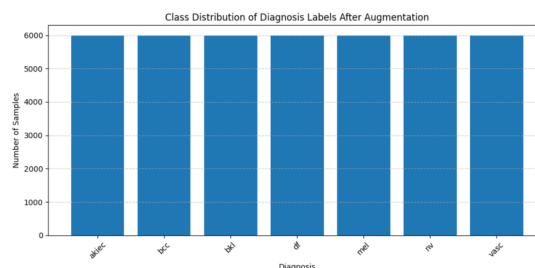


Figure 3: Distribution of skin lesion classes after augmentation.

Baseline Model

A baseline CNN was developed to establish an initial benchmark for skin cancer classification performance. The model serves as a simple yet effective foundation for comparison with more advanced architectures such as ResNet-18. It adopts an encoder-decoder structure, where the encoder includes three convolutional layers with increasing channel depths (32, 64, 128), each followed by batch normalization, ReLU activation, and max pooling. The decoder consists of a flattening layer and two fully connected layers that map extracted features to a seven-class probability distribution, as depicted in Figure 4. This lightweight architecture is designed for efficiency and minimal hyperparameter tuning while providing a reliable baseline for future improvements.

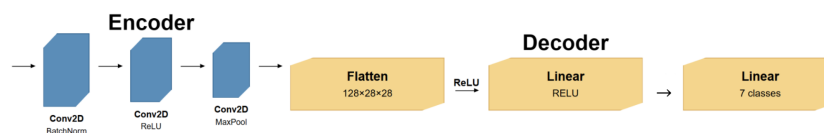


Figure 4: Architecture of the baseline CNN model.

Training this baseline CNN on the full HAM10000 dataset proved computationally intensive due to its large size. The process was exceptionally slow on both CPU and GPU, making full-scale training impractical within the project's time constraints. To make experimentation feasible, a smaller subset of 1,200 training and 300 validation images was used. Despite the limited data, the model demonstrated meaningful learning behavior, achieving a best training accuracy of 44.2% and a best validation accuracy of 67.0%, as shown in the accuracy plot in Figure 5.

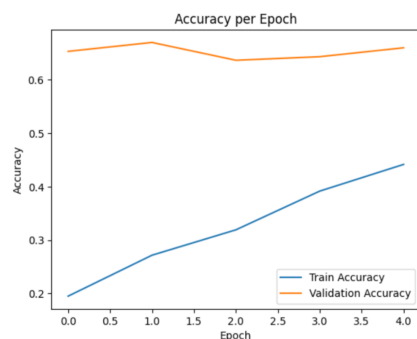


Figure 5: Accuracy curve of the baseline CNN model.

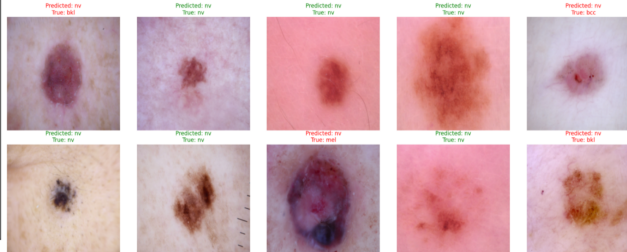


Figure 6: Predicted and true labels of the baseline CNN model.

The model also exhibited strong performance in identifying the nv class but struggled with other lesion types, as illustrated in Figure 6. This imbalance likely stems from the disproportionately large number of nv samples compared to other classes in the validation dataset. The relatively low training accuracy reflects the model's limited capacity to learn complex visual patterns from the small

dataset. Nevertheless, the stable validation accuracy indicates reasonable generalization without significant overfitting. These findings confirm the feasibility of CNN-based skin lesion classification and motivate the transition to a pretrained ResNet-18 architecture for improved performance.

Primary Model

Building on the baseline results, a transfer learning approach was implemented using a pretrained ResNet-18 architecture to enhance accuracy and generalization. ResNet-18 contains 18 layers and roughly 11.7 million parameters, providing moderate model complexity suitable for transfer learning on limited medical datasets. Its skip connections preserve gradient flow, enabling efficient feature extraction and stable optimization. This architecture was selected for its proven ImageNet performance and suitability for medical image analysis, offering a strong balance between complexity and computational efficiency. The model was customized by fine-tuning only the final residual block and fully connected layer while freezing earlier layers, replacing the original output with a seven-class classifier for the HAM10000 lesion categories. The dataset followed a 70-15-15 split for training, validation, and testing, using 5,000 training and 1,000 validation images augmented with random rotations, flips, and color jitter to improve generalization. Training used the Adam optimizer with a StepLR scheduler and early stopping to ensure stable convergence. The overall pipeline—from input images and augmentation through the pretrained ResNet-18 backbone, global average pooling, and modified classifier—is depicted in Figure 7.

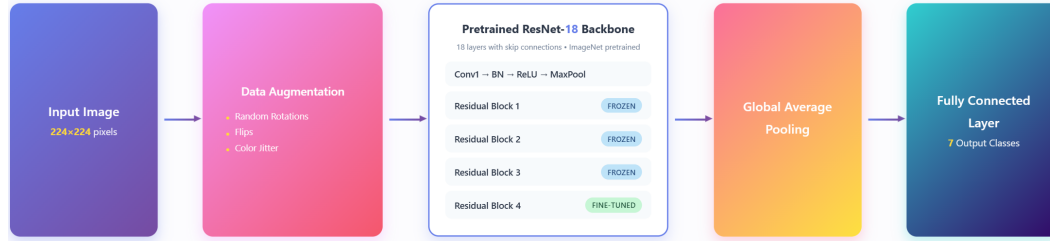


Figure 7: Architecture of the fine-tuned ResNet-18 model.

The model achieved best training and validation accuracies of 93.0% and 80.5%, respectively, as shown in the accuracy plot in Figure 8. Compared to the baseline CNN’s accuracy plot, the fine-tuned ResNet-18 exhibits a noticeably steeper and more stable improvement in both training and validation accuracy, reflecting faster convergence and more effective feature learning. The Grad-CAM visualization in Figure 9 further illustrates that the fine-tuned ResNet-18 model effectively focuses on the lesion region during prediction, indicating that the network learned to attend to clinically relevant features.

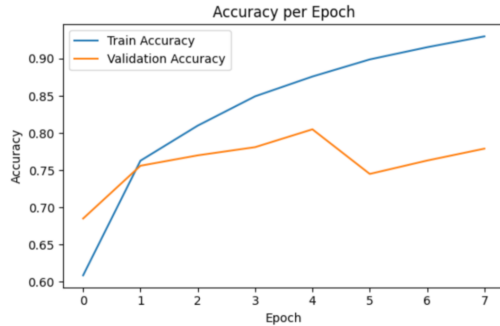


Figure 8: Accuracy curve of the fine-tuned ResNet-18 model.

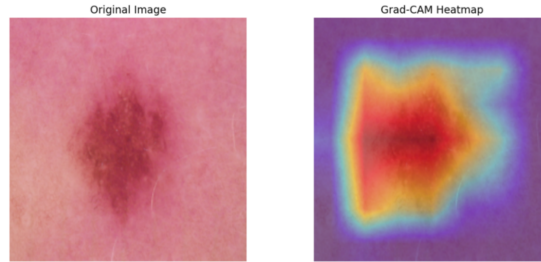


Figure 9: Grad-CAM visualization of the fine-tuned ResNet-18 model.

The fine-tuned ResNet-18 ultimately achieved substantially stronger and more stable performance than the baseline CNN’s 67.0% validation accuracy, confirming the effectiveness of pretrained feature extraction for skin-lesion classification. The primary limitation was the high computational cost of fine-tuning, which constrained large-scale experimentation and hyperparameter tuning within the project timeframe. Overall, these results affirm the ResNet-18 architecture as a robust foundation for automated skin-lesion classification and a solid basis for further optimization in the final model stage.

References

- [1] Canadian Skin Cancer Foundation, “Skin Cancer,” *Canadian Skin Cancer Foundation*. [Online]. Available: <https://www.canadianskincancerfoundation.com/skin-cancer/>. [Accessed: Sep. 24, 2025].
- [2] The Skin Cancer Foundation, “Skin Cancer Facts & Statistics,” *The Skin Cancer Foundation*. [Online]. Available: <https://www.skincancer.org/skin-cancer-information/skin-cancer-facts/>. [Accessed: Sep. 24, 2025].
- [3] R. Marks, “An overview of skin cancers: incidence and causation,” *The Lancet Oncology*, vol. 3, no. 4, pp. 165–169, 2002. [Online]. Available: <https://www.thelancet.com/journals/lanonc/article/PIIS1470204502006794/abstract>. [Accessed: Sep. 24, 2025].
- [4] Harvard Dataverse, “HAM10000 dataset,” *Harvard Dataverse*. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>. [Accessed: Sep. 24, 2025].
- [5] M. M. Musthafa, M. T. R., V. V. Kumar, and S. Guluwadi, “Enhanced skin cancer diagnosis using optimized CNN architecture and checkpoints for automated dermatological lesion classification,” *BMC Medical Imaging*, vol. 24, Art. no. 201, 2024. doi: 10.1186/s12880-024-01356-8. [Online]. Available: <https://bmcmmedimaging.biomedcentral.com/articles/10.1186/s12880-024-01356-8> (bmcmmedimaging.biomedcentral.com). [Accessed: Sep. 24, 2025].